Naive Bayes Classification - Quiz

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Kelas : A

1. Data

Day	Outlook	Temperature	Humidity	Wind	Play Ball
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Cloudy	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Cloudy	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

? Pertanyaan:

Berdasarkan data di atas, gunakan algoritma **Naive Bayes** untuk memprediksi apakah seseorang akan **Play = Yes atau No** jika kondisi cuaca berikut:

- Outlook = Sunny
- Temperature = Cool
- Humidity = High
- Windy = Weak

Petunjuk:

- Hitung probabilitas P(Yes) dan P(No)
- Hitung probabilitas kondisi fitur terhadap setiap kelas
- Gunakan Laplace smoothing jika perlu
- Tentukan prediksi akhir berdasarkan probabilitas tertinggi

Hitung Probabilitas Kelas

P(Yes) = 1/2
P(No) = 1/2

a. Outlook

Outlook	Yes(4)	No(4)
Sunny	0	3
Rain	2	1
Cloudy	2	0

b. Temperature

Temperature	Yes(4)	No(4)
Hot	1	2
Mild	1	1
Cool	2	1

c. Humidity

Humidity	Yes(4)	No(4)
High	2	3
Normal	2	1

d. Wind

Wind	Yes(4)	No(4)
Weak	3	2
Strong	1	2

Prediksi
Outlook = Sunny, Temperature = cool, Humidity = High, Wind = Weak
Hitung Probabilitas Kelas:
P(Yes|Fitur) = P(Yes) * P(Sunny|Yes) * P(Cool|Yes) * P(High|Yes) * P(Weak|Yes)

P(Yes) = 1/2
P(Sunny|Yes) 0 **Gunakan Laplace smoothing: (0+1)/(4+4)** 1/8
P(Cool|Yes) = 2/4
P(High|Yes) = 2/4
P(Weak|Yes) 3/4

P(Yes|Fitur) 0.01172

P(No|Fitur) = P(No) * P(Sunny|No) * P(Cool|No) * P(High|No) * P(Weak|No)

P(No) = 1/2
P(Sunny|No) = 3/4

P(Cool|No) =	1/4
P(High|No) =	3/4
P(Weak|No) =	1/2

P(No|Fitur)	0.03516

4. Menerapkan teorema Bayes

? Kesimpulan:

Karena P(No | Fitur) > P(Yes | Fitur), maka prediksi Naive Bayes untuk data:

Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Weak

adalah: "NO"